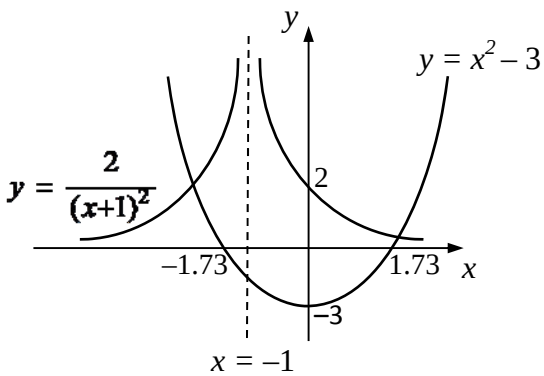




Suggested Mark Scheme

Qn	Key Steps
1	 x-coordinates of pts of intersection: -2.13 & 1.80 $x^2 \leq \frac{2}{(x+1)^2} + 3 \Rightarrow x^2 - 3 \leq \frac{2}{(x+1)^2}$ Solving, $-2.13 \leq x < -1$ or $-1 < x \leq 1.80$

Qn	Key Steps
2(a)	$\frac{1}{3}x^2 + ax + 3ab = -bx$ $\frac{1}{3}x^2 + (a+b)x + 3ab = 0$ Discriminant $= (a+b)^2 - 4\left(\frac{1}{3}\right)(3ab)$ $= a^2 + 2ab + b^2 - 4ab$ $= a^2 - 2ab + b^2$ $= (a-b)^2 > 0 \quad (\text{since } a \neq b \neq 0)$ $\therefore$ The equation has distinct real roots. (Shown)
2(b)	$(1-3x)^2 \leq 4$ $1-6x+9x^2 \leq 4$ $9x^2-6x-3 \leq 0$ $3x^2-2x-1 \leq 0$ $(3x+1)(x-1) \leq 0$ <u>Alternative method</u> $(1-3x)^2 \leq 4$ $[(1-3x)+2][(1-3x)-2] \leq 0$ $(3-3x)(-1-3x) \leq 0$ $(x-1)(1+3x) \leq 0$  $\therefore -\frac{1}{3} \leq x \leq 1$

Qn	Key Steps
3(a)	$\int \frac{(1-\sqrt{x})^2}{2\sqrt{x}} dx$ $= \int \frac{1-2\sqrt{x}+x}{2\sqrt{x}} dx$ $= \int \left(\frac{1}{2\sqrt{x}} - 1 + \frac{\sqrt{x}}{2} \right) dx$ $= x^{\frac{1}{2}} - x + \frac{x^{\frac{3}{2}}}{3} + c$
3(b)	Required area $= \int_2^9 (x-1)^{\frac{1}{3}} dx$ $= \left[\frac{(x-1)^{\frac{4}{3}}}{\frac{4}{3}} \right]_2^9$ $= \frac{3}{4} \left[(8)^{\frac{4}{3}} - 1 \right]$ $= \frac{45}{4}$

Qn	Key Steps
4(a)	$3e^x = 5 + 2e^{-x}$ $3e^{2x} - 5e^x - 2 = 0$ Let $y = e^x$, $3y^2 - 5y - 2 = 0 \quad \text{or} \quad 3(e^x)^2 - 5e^x - 2 = 0$ $(3y+1)(y-2) = 0$ $y = 2 \quad \text{or} \quad y = -\frac{1}{3}$ $e^x = -\frac{1}{3} \quad (\text{reject } \because e^x > 0 \quad \forall x \in \mathbb{R})$ $\therefore e^x = 2$ $x = \ln 2$
4(b)	$2 + 2\log_4(x-1) = \log_2 x$ $2 + \frac{2\log_2(x-1)}{\log_2 4} = \log_2 x$ $2 + \log_2(x-1) = \log_2 x$ $\log_2 x - \log_2(x-1) = 2$ $\log_2\left(\frac{x}{x-1}\right) = 2$ $\frac{x}{x-1} = 4$ $x = 4x - 4$ $x = \frac{4}{3}$

Qn	Key Steps								
5(a)	$y = \ln x - e^{2x} - x$ $\frac{dy}{dx} = \frac{1}{x} - 2e^{2x} - 1$ When $\frac{dy}{dx} = 0$, $\frac{1}{x} - 2e^{2x} - 1 = 0$ $x = 0.23727$ (Using GC) $y = -3.2831$ $(0.237, -3.28)$ is a stationary pt. <table><tr><td>x</td><td>0.23727^-</td><td>0.23727</td><td>0.23727^+</td></tr><tr><td>$\frac{dy}{dx}$</td><td>+ve</td><td>0</td><td>-ve</td></tr></table> <u>Alternate method</u> $\frac{d^2y}{dx^2} = -\frac{1}{x^2} - 4e^{2x} < 0$ for all x. $(0.237, -3.28)$ is a max pt. when $x = 1$, $y = -e^2 - 1$ $\frac{dy}{dx} = \frac{1}{1} - 2e^2 - 1 = -2e^2$ Equation of tangent at $x = 1$: $y - (-e^2 - 1) = (-2e^2)(x - 1)$ <u>Alternate method</u> $y = (-2e^2)x + c$ $-e^2 - 1 = -2e^2 + c$ $c = e^2 - 1$ Equation of tangent is $y = -2e^2x + e^2 - 1$ At $(-1, a)$, $a = -2e^2(-1) + e^2 - 1 = 3e^2 - 1$	x	0.23727^-	0.23727	0.23727^+	$\frac{dy}{dx}$	+ve	0	-ve
x	0.23727^-	0.23727	0.23727^+						
$\frac{dy}{dx}$	+ve	0	-ve						

5(b)

$$V = \frac{4}{3}\pi r^3$$

$$4.5\pi = \frac{4}{3}\pi r^3$$

$$r^3 = \frac{27}{8}$$

$$r = \frac{3}{2}$$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\frac{dA}{dt} = 8\pi r \frac{dr}{dt}$$

$$7 = 8\pi \left(\frac{3}{2}\right) \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{7}{12\pi} \text{ cm s}^{-1}$$

Qn	Key Steps
6 (i)	Let X be the length (in cm) of a randomly chosen fish in the pond. $X \sim N(25, \sigma^2)$ $P(X > 20) = 0.85$ Standardising, $P\left(Z > \frac{20 - 25}{\sigma}\right) = 0.85$ $\frac{-5}{\sigma} = -1.0364 \text{ (Using GC)}$ $\sigma = 4.8244 \approx 4.82 \text{ (3 s.f.) (Shown)}$
(ii)	Expected total price of 100 fish $= \$\{E[0.1(X_1 + X_2 + \dots + X_{100})]\}$ $= \$(0.1)(100)(25)$ $= \$250$
(iii)	$\therefore X \sim N(25, 4.8244^2)$ $P(X > 28) = 0.26702 = 0.267 \text{ (3 s.f.)}$
(iv)	P(exactly two fish are ready for the market) $= [P(X > 28)]^2 [P(X \leq 28)] \times 3$ $= [0.26702]^2 [1 - 0.26702] \times 3$ $= 0.157 \text{ (3 s.f.)}$

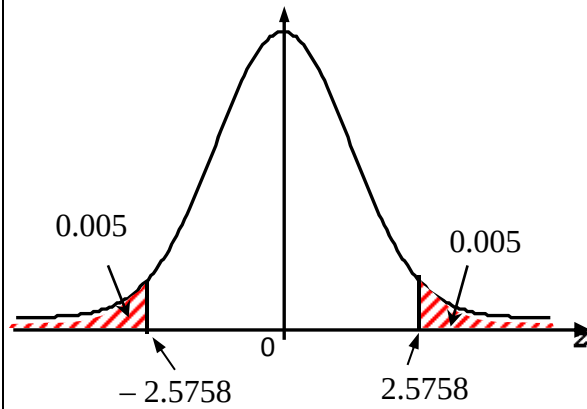
Qn	Key Steps
7(i)	$S \sim N(50, 3^2) \quad \& \quad W \sim N(140, 5^2)$ $W - 3S \sim N(140 - 3 \times 50, 5^2 + 3^2 \times 3^2)$ $\Rightarrow W - 3S \sim N(-10, 106)$ $P(W - 3S > 5) = 0.0726$
(ii)	Let $T = S_1 + S_2 + \dots + S_n \sim N(50n, 3^2 n)$ $P(T \leq 10000) > 0.99$ Standardising, $P\left(Z \leq \frac{10000 - 50n}{3\sqrt{n}}\right) > 0.99$ From GC, $\frac{10000 - 50n}{3\sqrt{n}} > 2.3263$ Solving using GC, $0 < n < 198.04, \quad n \in \mathbb{N}^+$ $\therefore$ greatest $n = 198$.

Qn	Key Steps
8(i)	Let X be the time taken (in minutes) for the new team to wash a randomly chosen car. $E(X) = 8.6$ $\text{Var}(X) = 1.5^2$ Mean of sample mean = 8.6 min Variance of sample mean = $\frac{1.5^2}{10} = 0.225 \text{ min}^2$
(ii)	After 6 months, let the new mean washing time for a car be a minutes. Since $n = 60$ is large, by Central Limit Theorem, $X_1 + X_2 + \dots + X_{60} \approx N(60a, 60(1.5^2))$ approx i.e. $X_1 + X_2 + \dots + X_{60} \approx N(60a, 135)$ approx $P(X_1 + X_2 + \dots + X_{60} < 402) = 0.6$ Standardising, $P\left(Z < \frac{402 - 60a}{\sqrt{135}}\right) = 0.6$ $\frac{402 - 60a}{\sqrt{135}} = 0.25335$ $a = 6.6509 = 6.65 \text{ (3 s.f.)}$

Qn	Key Steps
9(i)	$P(\text{at least 1 red ball is drawn}) = 1 - \frac{10}{15} \times \frac{9}{14} = \frac{4}{7}$ $\text{or } \frac{5}{15} \times \frac{10}{14} \times 2 + \frac{5}{15} \times \frac{4}{14}$ $= \frac{4}{7}$
(ii)	$P(\text{sum of numbers is 9}) = \frac{3}{15} \times \frac{3}{14} \times 2$ $= \frac{3}{35}$
(iii)	$P(\text{sum of numbers is 9} \text{at least 1 red ball is drawn})$ $= \frac{P(\text{sum of numbers is 9 \& at least 1 red ball is drawn})}{P(\text{at least 1 red ball is drawn})}$ $= \frac{P(4(R), 5(RGB)) + P(4(GB), 5(R))}{P(\text{at least 1 red ball is drawn})}$ $\text{or } \frac{P(4(R), 5(R)) + P(4(R), 5(GB)) + P(4(GB), 5(R))}{P(\text{at least 1 red ball is drawn})}$ $= \frac{2 \left(\frac{1}{15} \times \frac{3}{14} + \frac{2}{15} \times \frac{1}{14} \right)}{\frac{4}{7}}$ $\text{or } \frac{2 \left(\frac{1}{15} \times \frac{1}{14} + \frac{1}{15} \times \frac{2}{14} + \frac{2}{15} \times \frac{1}{14} \right)}{\frac{4}{7}}$ $= \frac{1}{12}$ Since $P(\text{sum of numbers is 9} \text{at least 1 red ball is drawn}) \neq P(\text{sum of numbers is 9})$, the 2 events are not independent.

10(a)	Obtain a staff list and group the staff into the following 3 strata: directors, managers & executives. <table><tr><td>No of directors</td><td>No of managers</td><td>No of executives</td></tr><tr><td>$5\% \times 60 = 3$</td><td>$20\% \times 60 = 12$</td><td>$75\% \times 60 = 45$</td></tr></table> Randomly select the number of staff from each stratum according to the numbers above to obtain a stratified sample. The advantage of stratified sampling is that the sample obtained is representative of the various strata (types of staff).	No of directors	No of managers	No of executives	$5\% \times 60 = 3$	$20\% \times 60 = 12$	$75\% \times 60 = 45$
No of directors	No of managers	No of executives					
$5\% \times 60 = 3$	$20\% \times 60 = 12$	$75\% \times 60 = 45$					
(b)(i)	Let X be the number of staff members who answers “yes” out of 60 staff members. $X \sim B\left(60, \frac{7}{9}\right)$ $\text{Var}(X) = 60 \times \frac{7}{9} \times \frac{2}{9} = \frac{280}{27}$ $\text{Standard deviation} = \sqrt{\frac{280}{27}} \text{ or } 3.22 \text{ (3s.f)}$						
(ii)	$E(X) = 60 \times \frac{7}{9} = \frac{140}{3} \text{ or } 46\frac{2}{3}$ $P\left(X > 46\frac{2}{3}\right) = 1 - P(X \leq 46) = 1 - 0.46787 = 0.53213 = 0.532 \text{ (3s.f)}$						
(iii)	From GC, $P(X = 46) = 0.11846$ $P(X = 47) = 0.12350$ $P(X = 48) = 0.11707$ $\therefore$ the most probable number is 47.						
(iv)	The answers are not valid as the probability of a staff member answering “yes” will not be approximately constant if the total number of staff members in the company is small.						

Qn	Key Steps
11(a)(i)	Unbiased estimate of the population mean $= \bar{x} = \frac{\sum x}{50} = \frac{270}{50} = 5.4$ Unbiased estimate of the population variance $= s^2 = \frac{1}{50-1} \sum (x - \bar{x})^2 = \frac{64.19}{49} = 1.31$
(ii)	Let X be the queuing time (in minutes) of a randomly chosen customer with population mean μ. To test $H_0 : \mu = 5$ against $H_1 : \mu > 5$ Perform a 1-tail test at 1% level of significance i.e. $\alpha = 0.01$ $\therefore n = 50$ is large, by Central Limit Theorem & under H_0, $\bar{X} \sim N\left(5, \frac{1.31}{50}\right)$ approx Using z-test, p-value = 0.00673 (using GC) Since p-value < 0.01, we reject H_0 and conclude that there is sufficient evidence at the 1% level of significance that Jerome's claim is valid.
(iii)	Not necessary $\therefore n = 50$ is large. By Central Limit Theorem, the distribution of the sample mean is approximately normal.
(b)	To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ Perform a 2-tail test at 1% level of significance $\therefore n = 50$ is large, by Central Limit Theorem, & under H_0, $\bar{X} \sim N\left(\mu_0, \frac{1.31}{50}\right)$ approx Standardising, $Z = \frac{\bar{X} - \mu_0}{\sqrt{1.31/50}} \sim N(0, 1)$ approx



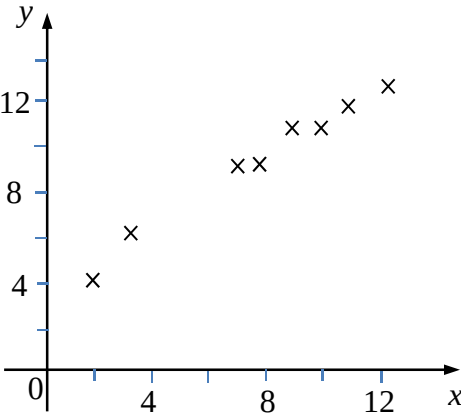
If the null hypothesis is rejected using z-test,

$$\frac{5.4 - \mu_0}{\sqrt{1.31/50}} < -2.5758 \quad \text{or} \quad \frac{5.4 - \mu_0}{\sqrt{1.31/50}} > 2.5758 \quad (\text{Using}$$

GC)

Set of values of μ_0

$$= \{ \mu_0 \in \mathbb{R} : \mu_0 < 4.98 \text{ or } \mu_0 > 5.82 \} \quad (3 \text{ s.f.})$$

Qn	Key Steps
12(i)	
(ii)	Product moment correlation coefficient, r $= 0.989$ (3 s.f.) (using GC) As r is close to 1, there is a (very) strong positive linear relationship between x and y.
(iii)	Equation of regression line of y on x: $y = 2.8743 + 0.83880x$ (5 s.f.) $y = 2.87 + 0.839x$ (3 s.f.) where $a = 2.87$ & $b = 0.839$
(iv)	$a = 2.87$ means it takes 2.87 min for a waiter to take Carl's order after he sat down at a table when there is no other customer in the restaurant. $b = 0.839$ means there is an increase of 0.839 min in time taken for a waiter to take Carl's order after he sat down at a table for every additional customer already seated in the restaurant.

(v)(a)	Time taken for a waiter to take Carl's order when 5 customers already seated in the restaurant $= 2.8743 + 0.83880(5)$ $= 7.07 \text{ min (3 s.f.)}$
(b)	Time taken for a waiter to take Carl's order when 20 customers already seated in the restaurant $= 2.8743 + 0.83880(20) = 19.7 \text{ min (3 s.f.)}$
(vi)	Estimating y given $x = 5$ is reliable as $r = 0.989$ is close to 1 and $x = 5$ is within the range of data given. i.e. $2 \leq x \leq 12$. Estimating y given $x = 20$ is unreliable as $x = 20$ is outside the range of data given where the linear relationship between x and y may not hold.